Simple View: Metric Lineage and Data Requirements

Power BI sample project | 9 October 2026

This document explains the 18 KPI tiles on the Simple View: what each metric represents, how it is calculated, which values can be ingested automatically, and what must be maintained manually or operationally.

> **Current-state caveat:** the semantic model in this project uses synthetic sample partitions. The APIs and source flows below describe the intended live-data implementation; they are not currently connected to this PBIP model.

How to read the source classifications

- **Automatic/API** means data can be extracted from an API or an automated CI output and then loaded into the model. It still requires credentials, scheduled ingestion, history retention, and correct mapping.
- **Manual/reference** means a business-owned mapping or assumption must be maintained.
- **Operational convention** means people or workflows must create and classify records consistently in a system before API ingestion can interpret them.

1. Are people using it?

Metric	Calculation and interpretation	Automatic/API source	Manual or operational dependency
Adoption Rate	Active Users divided by Assigned Seats. This is active usage as a share of assigned licences.	GitHub Copilot team metrics provide active users; the Copilot billing seats endpoint provides assigned seats.	Map GitHub teams to report teams/business units.
Active Users	The measure takes the peak `ActiveUsers` value per team over the selected period, then sums across teams. It is not a sum of daily distinct users.	GitHub Copilot team metrics endpoint, `GET /orgs/{org}/team/{team_slug}/copilot/metrics`.	Team slug and team-to-business-unit mapping.

Metric	Calculation and interpretation	Automatic/API source	Manual or operational dependency
Sustained Usage %	Sustained users divided by Active Users. A user is considered sustained after activity in at least three distinct weeks.	Copilot seats plus user-level Copilot metrics can produce assignment and active-week history.	Map users to teams. GitHub's metrics API only returns roughly 28 days, so ingest snapshots daily to preserve history and calculate weeks reliably.

2. Are they using more than autocomplete?

Metric	Calculation and interpretation	Automatic/API source	Manual or operational dependency
Code Acceptance Rate	Total suggestions accepted divided by total suggestions made.	GitHub Copilot metrics fields for code suggestions and code acceptances.	Team mapping. Validate metric aggregation against the Copilot admin UI.
Chat Sessions	Sum of daily chat values in `FactTeamDaily`.	GitHub Copilot metrics API chat breakdown.	Team mapping.
Agent Sessions	Sum of daily `AgentSessions` values in `FactTeamDaily`.	GitHub Copilot metrics API.	**Definition check required:** the guide's example populates this field from `copilot_dotcom_chat.total_engaged_users`, which is engaged users, not necessarily a session count. Verify the API field and rename/redefine the metric if needed before reporting it as sessions.

3. Is delivery improving?

Metric	Calculation and interpretation	Automatic/API source	Manual or operational dependency
Median PR Cycle Time	Median `PRCycleTimeHours` for rows with one or more merged PRs. The ingestion guide computes PR creation-to-merge elapsed time.	GitHub Pull Requests data, including created and merged timestamps; GraphQL is recommended in the GitHub-only guide.	Repository-to-team mapping and a consistent definition of which PRs count.
PRs Merged	Sum of merged PRs in the selected scope and period.	GitHub Pull Requests API/GraphQL, or the equivalent SDLC platform.	Repository-to-team mapping.

Metric	Calculation and interpretation	Automatic/API source	Manual or operational dependency
Estimated Hours Saved	Accepted suggestions × assumed minutes saved per suggestion ÷ 60.	Accepted-suggestion count from Copilot metrics.	A validated minutes-saved-per-suggestion assumption. The current DAX fallback is 0.8 minutes. This is an estimate, not directly measured time saved.

4. Is quality holding?

Metric	Calculation and interpretation	Automatic/API source	Manual or operational dependency
Change Failure Rate	Change failures divided by total changes. The GitHub-only guide treats an incident created within 24 hours of a successful production deployment as a change failure.	GitHub Deployments API joined to GitHub Issues.	Teams must consistently open incidents, apply the expected labels, and associate them with the affected repository. Agree and document the failure rule. Other organizations may use a different definition.
Test Coverage %	Average of monthly `TestCoveragePct` values in the quality fact table.	CI-generated coverage output, for example a GitHub Actions workflow publishing a JSON summary or artifact. This is not a Copilot API metric.	CI workflows must publish coverage and identify the repo/date correctly. Coverage may be blank until that workflow is in place.
MTTR	Average `MTTRHours` in the quality fact table; the GitHub-only guide derives elapsed time from incident creation to closure.	GitHub Issues timestamps in the GitHub-only design.	Operations must record and resolve incidents consistently. If Remedy or another service desk is authoritative, ingest that system instead.

5. Is value being realised?

Metric	Calculation and interpretation	Automatic/API source	Manual or operational dependency
Net Value	Gross Capacity Value + Avoided Defect Cost − Copilot Investment.	Depends on accepted suggestions, escaped-defect counts, seat counts, and credit usage.	Finance/engineering assumptions drive cost and capacity values. The current formula also hard-codes a baseline of **8 defects per team per month**; replace or validate that benchmark before using the value externally.
Capacity Released (FTEs)	Estimated Hours Saved × capacity-conversion percentage ÷ (hours per FTE month × number of selected months).	Accepted suggestions feed the estimated-hours-saved calculation.	Assumptions for minutes saved per suggestion, capacity-conversion percentage, and hours per FTE month. Current DAX fallbacks are **0.8 minutes**, **60%**, and **160 hours**.

Metric	Calculation and interpretation	Automatic/API source	Manual or operational dependency
ROI %	Net Value divided by Copilot Investment.	Inherits the data feeds used by Net Value and Investment.	Inherits all assumptions used in those measures; ROI is a modelled result, not a directly observed source value.

6. Are we managing risk and cost?

Metric	Calculation and interpretation	Automatic/API source	Manual or operational dependency
Inactive Seats	Assigned Seats minus Active Users.	Copilot billing seats plus Copilot usage metrics.	Team mapping; the result inherits the model's active-user peak logic.
Vulnerabilities Introduced	Sum of monthly vulnerability counts.	GitHub Advanced Security code-scanning alerts. Dependabot and secret-scanning alerts can be added for a broader security view.	Repository-to-team mapping. The guide warns that alert history is snapshot-based, so ingest regularly. Confirm whether the model's intended measure is newly introduced findings or currently open findings; the example endpoint filters open alerts.
Copilot Investment	Licence Cost + AI Credit Cost + Enablement Cost. Licence cost = seats × months × rate; AI cost = credits × rate; enablement = months × monthly cost.	Seat counts and AI-credit usage can come from GitHub APIs.	Assumptions for licence cost per seat/month, AI-credit cost per unit, and monthly enablement cost. Current DAX fallbacks are **R350**, **R0.15**, and **R45,000**. Use invoice/billing data if this tile should represent actual spend rather than estimated spend.

Other page inputs and benchmarks

- **Business Unit slicer:** needs a maintained organization mapping for business unit, portfolio/team, cost centre, eligible developer headcount, GitHub team slug, and repository ownership. The ingestion guide proposes a governance-repo `orgmap.csv`, with HR/Finance owning the organizational fields. Eligible headcount is not one of the 18 KPI tiles but the mapping supports correct grouping and filtering.
- **Date slicer:** uses the model's date table; it does not need an external operational feed.
- **Narrative targets:** the “What good looks like” statements are static report text, including adoption above 75%, sustained usage above 60%, acceptance above 30%, and coverage above 75%. The 42-hour PR-cycle baseline and 12% change-failure baseline are model/report constants, not API values. Review and govern these targets before presenting them as official benchmarks.
- **Data retention:** because Copilot metrics are limited to about 28 days and security alert history is snapshot-based, scheduled ingestion and retained Lakehouse history are important for trend reporting.

Project references

- Semantic calculations: `defaultProject.SemanticModel/definition/tables/_Measures.tmdl`
- Assumptions and current fallback values: `defaultProject.SemanticModel/definition/tables/Assumptions.tmdl`
- GitHub-only API, CI, and governance ingestion design: `docs/data-ingestion-guide.github.md`
- Hybrid GitHub/Azure DevOps source design: `docs/data-ingestion-guide.md`